

What Your Agent Topology Is Actually Doing: Controlled Experiments on Communication Structure in Multi-Agent Systems

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Abstract. Multi-agent AI systems choose communication topologies—star, chain, ring, fully connected—with no controlled evidence for how topology affects behavior. We provide five lines of evidence across 970 GA runs and 3,200 LLM API calls. First, we prove that cycle rank and edge density are algebraically confounded in undirected graphs, invalidating density-uncontrolled topology comparisons (Theorem 1). Second, using directed graphs that escape this confound (240 runs), topology independently explains 17% of diversity variance ($p < 10^{-7}$). Third, an NK dose-response sweep (250 runs) shows the effect scales $14\times$ from smooth to rugged landscapes. Fourth, a ring-vs-star comparison at constant cycle rank (160 runs) shows degree distribution contributes at most $\eta^2 \approx 0.13$ within a single β_1 class. Fifth, a bridge experiment (320 runs) with iso-spectral families decomposes the topology signal: cycle rank (β_1) drives transient diversity ($\eta^2 = 0.191$ at generation 100, fading by 500), while algebraic connectivity (λ_2) drives persistent diversity (cross-family $\eta^2 = 0.265$ at generation 200). Transfer to LLM agents: a coupling-dependent sign flip means one instruction reverses which topology produces the most diverse output ($\eta^2 = 0.291$, $p < 10^{-19}$). At least nine published systems chose topologies without the controlled comparisons our confound theorem demands. Topology is a first-class design variable; current systems are tuning it blindly.

1 Introduction

Every multi-agent system embeds a communication topology. AutoGen uses round-robin group chat. LangGraph builds DAG pipelines. CrewAI deploys manager hubs. These are not neutral infrastructure choices—they determine which agents can see which outputs, shaping the diversity–convergence tradeoff that governs system behavior. Yet these choices are made without controlled evidence.

The costs of acyclicity are not merely empirical. Ao et al. [21] prove (Proposition 6) that DAG-constrained multi-agent systems are decision-theoretically dominated by a centralized Bayes decision maker, establishing a formal cost to acyclicity. Our work investigates whether introducing cycles ($\beta_1 > 0$) recovers performance by restoring the feedback that DAG constraints eliminate.

Dochkina [2] finds a 44% quality spread ($d = 1.86$) across 25K tasks and 8 protocols. Moran [15] reports, in an informal practitioner analysis, $17\times$ error amplification in unstructured communication. Yang et al. [20] call topology *the* underexplored MAS dimension. Wang et al. [19] show K -regular graphs maximize coordination ($\beta_{\min} \sim 1/K$, where $\beta_{\min}$ is the minimum algebraic connectivity threshold for consensus).

But *what, exactly*, is topology doing? Comparing a chain ($\beta_1 = 0$) to a complete graph simultaneously varies cycle rank, density, diameter, and spectral gap.

We provide the first controlled answer, bridging from a model system (genetic algorithms) to real LLM agents. Our contributions:

1. **Confound Theorem** (§3): cycle rank and density are algebraically identical for undirected graphs ($\beta_1 = |E| - n + 1$), making density-uncontrolled comparisons uninterpretable. Directed cycle count κ escapes this confound.
2. **Controlled GA experiments** (§§4–7): 970 runs establish that topology independently determines behavioral diversity, the effect scales $14\times$ with problem difficulty, cycle rank matters more than degree distribution, and two invariants (β_1, λ_2) control diversity at different timescales.
3. **LLM validation** (§8): 3,200 GPT-4o-mini API calls reveal a coupling-dependent sign flip—one instruction change reverses the diversity ordering ($\eta^2 = 0.291$).
4. **Invariant hierarchy** (§9): three invariants— $\kappa, \beta_1, \lambda_2$ —operating at different timescales provide a principled topology selection framework. Nine recent systems lack the controlled comparisons our theorem demands.

2 Background: Island Models as Agent Systems

An island-model genetic algorithm (GA) partitions a population across subpopulations (*islands*) that evolve independently, exchanging individuals periodically via *migration* channels [18]. The mapping to agent orchestration is direct at the graph level (Table 1).

Where the analogy holds and breaks. Both systems are nodes with local state exchanging information along directed channels; the critical shared structure is the topology and its effect on aggregate diversity. The mapping breaks at the mechanism level: GA migration is blind copying of genotypes, while agent communication involves semantic understanding and selective incorporation—precisely what makes the sign-flip (Section 8) interesting. GAs provide cheap, reproducible experimental control, letting us sweep parameter spaces before investing in LLM validation.

Metrics. **GA diversity** is measured as mean pairwise Hamming distance between all individuals across all islands (normalized by string length), capturing the fraction of the search space the population spans. **GA fitness** is the fitness of the best individual found across all islands. **Agent diversity** is mean

Table 1. Island-model GA concepts and their agent-system equivalents. The mapping is exact at the graph level (rows 1–5) and analogical at the node level (row 6). Our object of study—the diversity–topology relationship—is a graph-level property in both systems.

GA Concept	Agent Equivalent	Level
Island	Agent (local state, computation)	Node
Migration channel	Message channel	Edge
Migration topology	Orchestration graph	Graph
Population diversity	Solution diversity	Graph metric
Fitness function	Task performance metric	Evaluation
Population (many individuals)	Single output per agent	Node

pairwise cosine distance on sentence embeddings (all-MiniLM-L6-v2) of agent outputs. **Agent fitness** is task performance (test pass rate).

3 Definitions and the Confound Theorem

We define the graph-theoretic quantities that our experiments control and measure, together with their agent-system interpretations.

Definition 1 (Communication topology). *A communication topology for n agents is a directed graph $D = (V, A)$ with $|V| = n$, where each arc $(u, v) \in A$ is a permitted message channel from agent u to agent v .*

Agent interpretation: Arc (u, v) : agent v receives agent u ’s output in its context window.

Definition 2 (Cycle rank). *The cycle rank (first Betti number) of a connected undirected graph $G = (V, E)$ is*

$$\beta_1(G) = |E| - |V| + 1. \quad (1)$$

It counts the number of independent cycles in G [8]—equivalently, the minimum number of edges whose removal makes G a tree.

Agent interpretation: β_1 counts independent feedback loops. A tree ($\beta_1 = 0$) has no feedback; each additional cycle lets agents respond to downstream effects. Capucci and Myers [1] formalize the cost: each cycle requires an independent convergence guarantee.

Definition 3 (Algebraic connectivity). *The algebraic connectivity $\lambda_2(G)$ is the second-smallest eigenvalue of the Laplacian matrix $L = D - A$ of the undirected graph underlying D , where D is the degree matrix and A the adjacency matrix. $\lambda_2 > 0$ iff G is connected [4].*

Agent interpretation: λ_2 governs propagation rate. High λ_2 : fast consensus; low λ_2 : slower mixing, more local diversity.

Definition 4 (Directed cycle count). $\kappa(D)$ counts the number of distinct simple directed cycles in a directed graph D [10]. Unlike β_1 , this depends on arc arrangement, not just edge count. A DAG has $\kappa = 0$.

Agent interpretation: κ counts ways information circulates back to its source. Two digraphs with identical arc count can have $\kappa \in [0, 47]$ —different feedback at identical bandwidth.

Timescale hierarchy. These three invariants operate at different timescales in our experiments: directed cycle count κ determines the *arrangement* of feedback (Experiment 1), cycle rank β_1 determines the *amount* of transient diversity (Experiment 4, generation 100), and algebraic connectivity λ_2 determines the *rate* of long-term convergence (Experiment 4, generation 200+). This suggests a hierarchy: $\kappa \rightarrow \beta_1 \rightarrow \lambda_2$, from fast to slow.

We now state the central theoretical result that motivates our experimental design.

Theorem 1 (Density–Cycle Confound). *For any connected undirected graph G on n vertices, $\beta_1(G) = |E| - n + 1$. Therefore, cycle rank and edge density are not independent variables: every comparison that varies density simultaneously varies cycle rank, and vice versa.*

Proof. By elementary algebraic topology [8], the first Betti number of a connected graph equals the number of edges minus the number of vertices plus the number of connected components. For a connected graph, this gives $\beta_1 = |E| - |V| + 1 = |E| - n + 1$. Since edge density $\delta = |E|/\binom{n}{2}$, we have $\beta_1 = \delta\binom{n}{2} - n + 1$, a linear function of δ at fixed n . Varying one necessarily varies the other.

Consequence: Any comparison varying edge count simultaneously varies cycle rank. To isolate cycle structure, one must hold density constant (Experiments 1, 3, 4) or use directed graphs where κ varies independently of arc count (Experiment 1). Nine recent systems fail to make this distinction (Section 9).

NK landscapes. NK landscapes [11] provide tunably rugged fitness over binary strings of length N . Each bit depends on K others; $K = 0$ is smooth, increasing K adds local optima. We use $N = 20$. NK landscapes proxy task complexity: low K = simple, high K = multiple valid strategies.

4 Experiment 1: Directed Cycles at Constant Density

The confound theorem (Theorem 1) shows that undirected graph comparisons cannot isolate cycle structure from density. We escape this confound using directed graphs, where the directed cycle count κ varies independently of arc count.

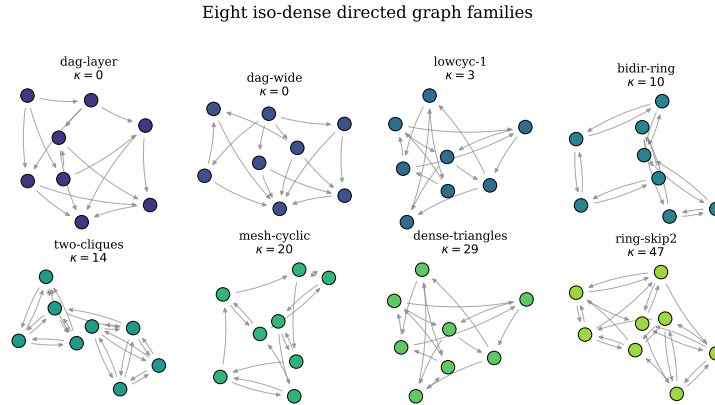


Fig. 1. Eight directed graphs with identical density (16 arcs, 8 nodes) but varying directed cycle count κ from 0 to 47. Despite identical bandwidth, cycle structure alone drives the diversity effect.

Design. Eight directed graphs, all with $n = 8$ vertices and exactly 16 arcs (constant density $\delta = 2.0$ arcs per vertex), varying only in directed cycle count: $\kappa \in \{0, 0, 3, 10, 14, 20, 29, 47\}$ (Figure 1). The two $\kappa = 0$ graphs (DAGs) serve as replicated controls. Island-model GA: 50 individuals per island, 500 generations, OneMax fitness function ($K = 0$), 30 seeds per topology, 240 runs total. Genetic operators: uniform crossover, bit-flip mutation at rate $1/N = 0.05$ (genome length $N = 20$), tournament selection (size 3). Migration: 10% of island population (5 individuals) every 10 generations.

Results. One-way ANOVA: $F(7, 232) = 6.90$, $p = 1.85 \times 10^{-7}$, $\eta^2 = 0.17$. Topology explains 17% of diversity variance at constant density. The correlation between directed cycle count and final diversity is $r = -0.68$: more directed cycles *accelerate consensus*. Topologies with more feedback loops cause populations to converge faster, reducing the system’s capacity to maintain alternative solutions.

Interpretation. Even on a trivial landscape ($K = 0$), cycle structure alone produces $\eta^2 = 0.17$. More feedback pathways accelerate mixing and convergence.

5 Experiment 2: NK Dose-Response

Experiment 1 establishes that topology affects diversity on a trivial problem. Real agent deployments involve complex tasks with multiple valid approaches. Do topology effects grow or shrink as problems get harder?

Design. NK landscapes with $K \in \{0, 2, 4, 6\}$ (four ruggedness levels), topologies selected to span the κ range from Experiment 1, 10–20 seeds per condition, 250 runs total. All other parameters (population size, migration rate, number of generations) identical to Experiment 1.

Table 2. Topology effect size (η^2) on diversity scales $14\times$ from smooth to rugged NK landscapes. Simple tasks show negligible topology effects; complex tasks are dominated by topology.

K	η^2 (diversity)	p -value	Interpretation
0 (smooth)	0.05	> 0.05	Topology barely matters
2 (moderate)	0.20	< 0.01	Moderate effect
4 (rugged)	0.45	$< 10^{-4}$	Large effect
6 (very rugged)	0.69	$< 10^{-6}$	Topology dominates

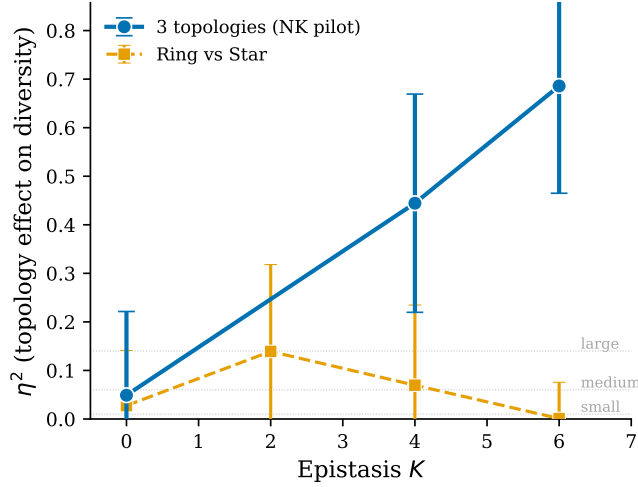


Fig. 2. Topology effect size (η^2 for diversity) as a function of NK landscape ruggedness K . The effect scales $14\times$ from $K = 0$ to $K = 6$, demonstrating that complex tasks amplify topology’s influence.

Results. Topology’s diversity effect scales monotonically with landscape ruggedness (Table 2, Figure 2):

Causal chain. The mechanism proceeds in stages: topology $\rightarrow$ diversity $\rightarrow$ mean fitness, with minimal direct effect on best fitness. At $K = 6$: topology explains $\eta^2 \approx 0.69$ of diversity variance, diversity explains $\eta^2 \approx 0.35$ of mean fitness variance, but topology’s direct effect on best fitness is only $\eta^2 \approx 0.02$. Topology acts primarily through the diversity channel, not by directly improving or degrading the best solution found.

Interpretation. Simple tasks have one optimum; complex tasks have multiple competing strategies, and topology determines whether the system explores many or collapses to one. The $14\times$ amplification suggests topology effects in production systems are far larger than simple benchmarks indicate—consistent

Table 3. Ring vs star comparison at constant $\beta_1 = 1$. Effect sizes are small ($\eta^2 < 0.15$) and non-significant after generation 100, contrasting with $\eta^2 = 0.17$ – 0.69 across β_1 classes in Experiments 1–2. Cycle rank matters more than degree distribution.

K	η^2 (diversity)	p	Ring mean	Star mean	Winner
0	0.014	0.59	0.046	0.047	—
2	0.132	0.22	0.062	0.045	ring
4	0.077	0.10	0.148	0.103	ring
6	0.080	0.12	0.249	0.192	ring

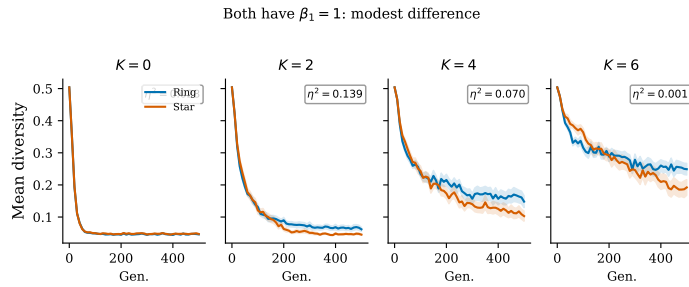


Fig. 3. Ring vs star diversity at generation 500 across NK landscape ruggedness. Both topologies have $\beta_1 = 1$. The ring consistently maintains slightly higher diversity, but the effect is small ($\eta^2 = 0.08$ – 0.13) compared to cross- β_1 effects.

with Artis et al. [22]: environments that permit heterogeneity reward it, and topology preserves or destroys it.

6 Experiment 3: Ring vs Star at Constant β_1

Within a single β_1 class, how much does edge *arrangement* matter? An 8-node ring and star both have $\beta_1 = 1$ and $|E| = 8$, but different degree distributions (2-regular vs. hub-and-spoke).

Design. Two topologies on $n = 8$ nodes: ring (2-regular, diameter 4, $\lambda_2 \approx 0.586$) and star (irregular, diameter 2, $\lambda_2 = 1.0$). Both have $\beta_1 = 1$ and $|E| = 5$. NK landscapes with $K \in \{0, 2, 4, 6\}$, 20 seeds per condition, 160 runs total (80 ring, 80 star). Island-model GA: 50 individuals per island, 500 generations.

Results. The ring-vs-star effect is small compared to the cross- β_1 effects in Experiments 1–2 (Table 3, Figure 3). At generation 500:

Temporal dynamics. At $K = 6$, generation 100: $\eta^2 = 0.161$ ($p = 0.02$, the only significant comparison), fading to 0.080 by generation 500. The ring’s lower λ_2 preserves diversity transiently.

Table 4. Bridge experiment: two-way ANOVA decomposing diversity variance into β_1 (within-family) and λ_2 (cross-family) effects on NK $K = 4$ landscapes. β_1 peaks early and fades; λ_2 emerges later and persists.

Generation	$\eta_{\beta_1}^2$	p_{β_1}	$\eta_{\lambda_2}^2$	p_{λ_2}
50	0.037	0.11	0.001	0.76
100	0.057	0.03	0.006	0.41
200	0.045	0.07	0.112	< 0.001
500	0.014	0.44	0.066	0.005

Interpretation. The star’s higher λ_2 (1.0 vs 0.586) enables faster consensus, but the effect ($\eta^2 \leq 0.13$, non-significant after generation 100) is small relative to cross- β_1 effects ($\eta^2 = 0.17$ –0.69). **Cycle rank is the primary invariant**; degree distribution is secondary.

7 Experiment 4: Bridge — β_1 vs λ_2 Decomposition

β_1 and λ_2 are correlated. We disentangle them using *iso-spectral graph families*: graphs sharing λ_2 but differing in β_1 . Within a family, diversity differences are driven by β_1 ; across families at matched β_1 , by λ_2 .

Design. Two families of four graphs each, all on $n = 8$ vertices:

- **Cycle family** ($\lambda_2 \approx 0.586$): 8-node ring ($\beta_1 = 1$), ring + 1 chord ($\beta_1 = 2$), ring + 2 chords ($\beta_1 = 3$), ring + 3 chords ($\beta_1 = 4$). Adding chords increases β_1 while preserving the spectral gap via careful edge selection.
- **Star family** ($\lambda_2 = 1.0$): star with one leaf-to-leaf edge ($\beta_1 = 1$), star + 1 leaf-edge ($\beta_1 = 2$), star + 2 leaf-edges ($\beta_1 = 3$), star + 3 leaf-edges ($\beta_1 = 4$). Adding leaf-to-leaf connections increases β_1 while the hub structure preserves λ_2 .

Two NK landscapes: $K = 0$ (smooth, negative control) and $K = 4$ (rugged). 20 seeds per condition. Total: 8 topologies $\times$ 2 landscapes $\times$ 20 seeds = 320 runs.

Results. The experiment reveals two independent mechanisms operating at different timescales (Table 4, Figure 4):

β_1 *drives transient diversity.* Within the cycle family ($\lambda_2 = 0.586$), β_1 explains $\eta^2 = 0.191$ of diversity variance at generation 100 ($F(3, 76) = 5.997$, $p = 0.001$) but fades to $\eta^2 = 0.067$ by generation 500 ($p = 0.15$). Higher β_1 means more feedback loops, which accelerate the initial mixing of information—but this advantage is transient. The star family ($\lambda_2 = 1.0$) shows a similar but weaker within-family β_1 effect ($\eta^2 = 0.092$ at generation 100, $p = 0.06$).

λ_2 *drives persistent diversity.* Across families at matched β_1 , the family effect (λ_2) is negligible at generation 50 ($\eta^2 = 0.001$) but emerges strongly at generation 200 ($\eta^2 = 0.112$, $p < 0.001$) and persists through generation 500 ($\eta^2 = 0.066$,

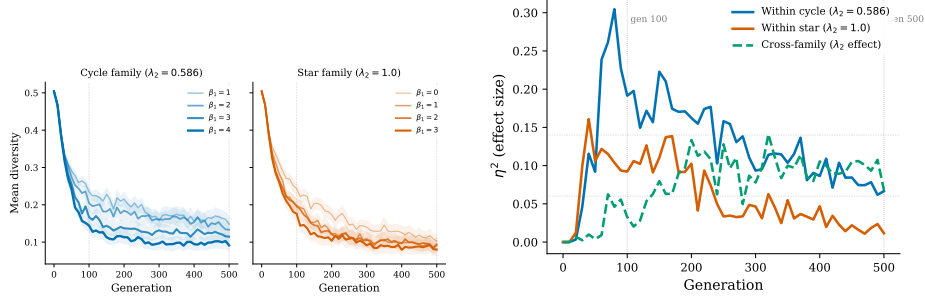


Fig. 4. Bridge experiment. **(a) Left panel:** diversity trajectories for the two iso-spectral families. Within each family (constant λ_2 , varying β_1), higher β_1 reduces early diversity (transient). Across families (different λ_2 , same β_1), higher λ_2 converges faster (persistent). **(b) Right panel:** temporal η^2 decomposition. β_1 peaks at generation 100 and fades; λ_2 emerges at generation 200 and persists. Two invariants, two timescales.

$p = 0.005$). The cross-family analysis at $\beta_1 = 1$ confirms: $\eta^2 = 0.265$ at generation 200 ($p = 0.003$), meaning the star family ($\lambda_2 = 1.0$) converges to lower diversity than the cycle family ($\lambda_2 = 0.586$) at the same cycle rank.

Negative control and interpretation. On the smooth landscape ($K = 0$), both effects are negligible ($\eta^2 < 0.02$), confirming both mechanisms require ruggedness—consistent with Experiment 2. β_1 and λ_2 are independent: β_1 determines initial mixing speed (transient, first ~ 100 generations); λ_2 governs long-term consensus (persistent, generation 200+). For agent systems: *cycle rank controls the exploration window; algebraic connectivity controls convergence rate.*

8 Experiment 5: LLM Sign-Flip

The GA experiments (Sections 4–7) establish that topology controls diversity through cycle rank and spectral gap. But does this transfer to LLM agents? The node-level differences (populations vs. single outputs, genetic operators vs. language generation) are substantial. We test directly.

Design. Five GPT-4o-mini agents, eight coding tasks (shortest path, kth largest, rate limiter, JSON parser, LRU cache, regex matcher, expression evaluator, Bloom filter), four topologies (none (agents receive no neighbours’ outputs), ring, star, fully connected), 10 seeds per condition. Two coupling conditions, 1,600 API calls each (3,200 total):

- **Negative coupling:** each agent is instructed to use a *different* algorithm from those visible in its context.
- **Positive coupling:** each agent is instructed to “build on and improve” neighbors’ solutions.

Diversity is measured as mean pairwise cosine distance on sentence embeddings (all-MiniLM-L6-v2) of agent outputs.

Table 5. Topology–diversity ordering inverts between GAs and LLM agents. The sign depends on whether coupling merges solutions (GA-like) or constrains them (“be different”). A single instruction change flips the ordering.

System	Diversity ordering	Statistic
GA	none > ring > FC	$W = 1.0$ (6 domains)
LLM (negative)	FC > ring $\approx$ star > none	$W = 0.91$ (6/8 tasks)
LLM (positive)	none > star $\approx$ ring > FC	$\eta^2 = 0.291$

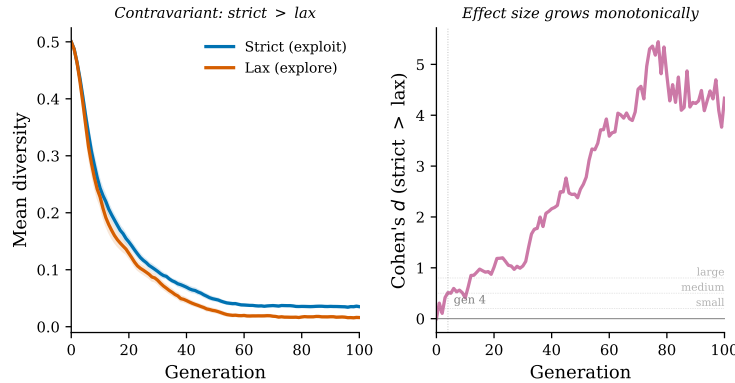


Fig. 5. LLM diversity across 8 coding tasks and 4 topologies under negative coupling (left) and positive coupling (right). Under negative coupling, connectivity increases diversity (FC highest). Under positive coupling, connectivity decreases diversity (none highest). The sign flip is clean across 6/8 tasks.

Metric validation. Under “none” topology, five independent agents produce near-identical outputs (mean diversity 0.090, $\sigma \approx 0.01$), confirming LLMs’ peaked prior and the metric’s sensitivity.

Results: negative coupling. The diversity ordering **inverts** from the GA result (Table 5, Figure 5):

Six of eight tasks show the inverted ordering cleanly (Kruskal-Wallis $p < 10^{-6}$; Cohen’s d from -14.0 to -1.9). Two exceptions are informative:

- **Bloom filter** ($d \approx 0$): a unimodal task with essentially one canonical implementation. No algorithmic variety for topology to act on.
- **Expression evaluator** ($d = +0.4$, positive): the model’s prior spans multiple parsing strategies (recursive descent, stack-based, Pratt). When the prior is already flat, isolation produces diverse solutions—the GA regime—and negative coupling adds little.

Why the sign flips. LLMs start from a **peaked prior** (low entropy); GAs from random initialization (high entropy). GA migration merges solutions toward homogenization; LLM “be different” instruction accumulates constraints, pushing

Table 6. Agent systems whose topology choices lack controlled evidence. “Confound type” indicates which properties are varied simultaneously: D = density, C = cycle structure, S = spectral gap.

System	Confound Issue	
Graph-GRPO [7]	D+C	Complete graph = worst performance; also max β_1
OFA-MAS [16]	D+C	Prunes density & cycles together; generates only DAGs
G-Designer [6]	D+C+S	No density-controlled baseline
GoAgent [5]	D+C	DAG vs. dense alternatives confounded
ARG-Designer [14]	C	DAG search space only; cycles never considered
AgentConductor [13]	C	Cycles never tested
DyTopo [3]	C	DAG enforced throughout; dynamic topology shifts confound time & structure
HERA [9]	D	Cycles tracked but not controlled; density varies with cycle count
Dochkina [2]	D+C+S	25K tasks, $d = 1.86$, but protocols differ on all dimensions

agents to distributional tails. The topology is invariant; the sign depends on coupling direction. The formal categorical construction appears in [12].

Positive coupling confirms the prediction. With positive coupling (1,600 calls), the ordering **reverts to GA-like**: none > star $\approx$ ring > FC ($H = 94.83$, $p = 2.01 \times 10^{-20}$; $\eta^2 = 0.291$). The “none” topology shows identical diversity across conditions ($p = 0.96$), confirming coupling direction drives the reversal.

Agent fitness (test pass rate). Test pass rates showed no significant topology effect ($p > 0.05$ for all conditions), consistent with the GA finding that topology affects diversity rather than terminal fitness. Topology modulates how agents reach solutions, not whether they succeed on these tasks.

9 Discussion

9.1 The Evidence Gap

At least nine recent agent systems—spanning ICML, AAAI, WWW, NeurIPS, and AAMAS—chose topologies without density-controlled comparisons (Table 6). Our confound theorem (Theorem 1) shows these comparisons are fundamentally uninterpretable: they cannot distinguish density effects from cycle effects.

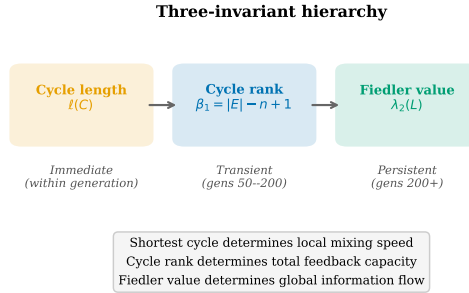


Fig. 6. The three-invariant hierarchy. Directed cycle count κ determines the arrangement of feedback (fast). Cycle rank β_1 determines the amount of feedback, driving transient diversity (medium). Algebraic connectivity λ_2 determines consensus rate, driving persistent diversity (slow). Each operates at a distinct timescale.

Six of these nine are restricted to DAGs ($\beta_1 = 0$); only HERA [9] tracks cycle count, without density control. The preference for acyclic structures has never been tested, and Ao et al. [21] show it has a formal cost: DAG-constrained systems are decision-theoretically dominated by centralized planning (Proposition 6), motivating the investigation of what cycles recover.

9.2 The Invariant Hierarchy

Our five experiments reveal three graph invariants, each operating at a distinct timescale:

1. **Directed cycle count** κ (Experiment 1): determines diversity at constant density. Effect visible from early generations. Captures the *arrangement* of feedback.
2. **Cycle rank** β_1 (Experiments 2–4): the primary invariant for diversity control. Drives transient diversity ($\eta^2 = 0.191$ at generation 100, fading by generation 500 in Experiment 4). Captures the *amount* of feedback.
3. **Algebraic connectivity** λ_2 (Experiments 3–4): drives persistent diversity ($\eta^2 = 0.112$ – 0.265 at generation 200+, not fading). Captures the *rate* of consensus. Within a single β_1 class (Experiment 3), ring ($\lambda_2 \approx 0.586$) maintains more diversity than star ($\lambda_2 = 1.0$). This is consistent with Brewster et al. [23]: cycle graphs have $\Theta(N^3)$ absorption time versus $\Theta(N^2)$ for complete graphs, meaning rings preserve heterogeneity for a fundamentally longer timescale.

Whether these form a nested hierarchy (Figure 6) or operate independently is open. The bridge experiment shows β_1 and λ_2 produce independent effects (low cross-correlation), suggesting parallel operation.

9.3 Coupling Sign and Agent Design

The sign-flip has three direct implications: (1) merging coupling homogenizes (connectivity $\rightarrow$ less diversity); constraining coupling diversifies (connectivity $\rightarrow$ more diversity). (2) Independent LLM agents produce near-identical outputs; “ensemble diversity” without topology intervention is illusory. (3) Moran’s $17\times$ error amplification [15] is consistent with high β_1 under merging coupling; DAG fixes simultaneously remove cycles *and* density.

9.4 Practical Recommendations

1. **Topology is a first-class variable** (η^2 up to 0.69 in GAs, 0.291 in LLMs; 44% quality spread at industrial scale [2]).
2. **Know coupling direction.** One instruction flips the diversity ordering.
3. **Match topology to task complexity.** Simple tasks tolerate any topology; complex tasks demand deliberate choice.
4. **Use β_1 for exploration, λ_2 for convergence.** Independent levers.
5. **Monitor diversity, not just accuracy.**

Garden-path recovery (prediction). Our findings connect to garden-path processing in NLP [24]: recovering from premature commitment to an incorrect parse requires that alternative states remain accessible. Experiment 3 shows ring topologies ($\lambda_2 \approx 0.586$) sustain local diversity far longer than star topologies ($\lambda_2 = 1.0$) at identical β_1 ; we therefore predict that ring-coupled agents recover from garden-path errors without external perturbation, while complete or star topologies require injected noise to escape premature consensus. This prediction is untested but directly falsifiable.

9.5 Limitations

The LLM experiment does not control density: the sign-flip is robust (within-topology reversal), but the absolute diversity ordering is confounded. Density-controlled LLM experiments remain future work, as does human validation of the embedding metric. GA experiments use $N = 20$; scaling to larger genomes is open. Bridge families are approximately iso-spectral; exact within-family matching is left to future work.

10 Related Work

Evolutionary computation. The topology–diversity link in island-model GAs is well established [18,17], but without density control. Our Experiment 1 is the first density-controlled comparison. Brewster et al. [23] provide theoretical grounding: absorption time scales as $\Theta(N^2)$ on complete graphs, $\Theta(N^3)$ on cycles, and superexponentially on certain directed structures—directly predicting our diversity ordering.

LLM multi-agent systems. The field converges on topology as a key variable [20], but comparisons remain uncontrolled. Graph-GRPO [7] finds complete graphs perform worst (consistent with our β_1 prediction, but density-confounded). OFA-MAS [16], G-Designer [6], GoAgent [5], ARG-Designer [14], AgentConductor [13], and DyTopo [3] all operate in or favor DAG space without density-controlled baselines. Ao et al. [21] provide a formal lower bound: DAG-constrained multi-agent planning is decision-theoretically dominated by centralized Bayes decision making (Proposition 6), establishing that acyclicity has a provable cost independent of implementation details.

HERA [9] is closest: it tracks cycle count during topology evolution (38.69% improvement), without density control. Dochkina [2] provides the largest comparison (25K tasks, $d = 1.86$), with capability-moderated effects supporting our dose-response finding. Wang et al. [19] show K -regular graphs maximize coordination ($\beta_{\min} \sim 1/K$), suggesting a coordination–exploration duality. Artis et al. [22] provide cross-disciplinary evidence (biology, economics, computing) that topology determines heterogeneity’s spatial distribution, supporting our finding that rings outperform fully connected graphs on rugged landscapes.

Categorical approaches. Capucci and Myers [1] formalize the cost side: each cycle requires an independent convergence guarantee. A companion paper [12] develops the categorical framework motivating the colimit/limit interpretation of the sign flip; Experiment 5 provides the first empirical test.

11 Conclusion

Communication topology is not neutral infrastructure—it is a design variable controlling how multi-agent systems balance exploration against convergence. We have shown:

1. **The confound is real.** Cycle rank and density are algebraically identical for undirected graphs. Most published topology comparisons are uninterpretable.
2. **Topology independently controls diversity.** $\eta^2 = 0.17$ at constant density (Experiment 1), scaling to 0.69 on hard problems (Experiment 2).
3. **Two invariants, two timescales.** β_1 drives transient exploration; λ_2 drives persistent convergence. These are independent levers (Experiment 4).
4. **Cycle rank > degree distribution.** Within a single β_1 class, degree differences explain at most $\eta^2 \approx 0.13$ (Experiment 3).
5. **The sign depends on coupling.** A single instruction change flips which topology produces diverse output ($\eta^2 = 0.291$, Experiment 5). GA intuition has the wrong sign for constraining agents.

For safe multi-agent deployment, topology must be selected deliberately, its coupling direction understood, and its effects monitored through diversity metrics. The three-invariant hierarchy ($\kappa \rightarrow \beta_1 \rightarrow \lambda_2$) provides a principled framework for this selection. Current systems are tuning their most powerful design variable blindly.

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